

Broadband tunable sound transparency via underwater metasurfaces

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Sound waves have consistently served as vital carriers of information in the physical world. However, in the realms of hydroacoustics and medical ultrasound, the impedance mismatch between the solid and liquid poses significant challenges, resulting in energy dissipation that hinders the effective transmission of information. Hence, achieving underwater sound transparency holds tremendous significance for advancing underwater acoustic communication technology and fostering innovations in the research and development of medical ultrasound equipment. This research endeavor leverages multimode coupling to transcend the limitations of previous narrow-band sound transparency designs. Confirmed by experiments, this work achieves broadband sound transparency spanning from 170 to 230 kHz. Furthermore, leveraging the abundance of geometric parameters and the flexibility of degrees of freedom, we introduce a dual broadband design, which exhibits dual broadband sound transparency with passbands of 280–350 kHz and 450–510 kHz. Crucially, our research serves as a stepping stone towards realizing broadband acoustic underwater metasurfaces, thereby paving the way for advancements in ultrasonic imaging and underwater acoustic communications.

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I. INTRODUCTION

Sound waves serve as one of the primary media for information transmission. Nevertheless, due to significant differences in impedance among propagation media, sound waves encounter various obstacles during propagation. Sound transparency enables the adjustment of impedance relationships in complex media, allowing sound waves to propagate more smoothly while reducing reflections and scattering at interfaces. This, in turn, enhances the propagation efficiency and energy utilization of sound waves. In fields such as medical ultrasound, underwater detection, and underwater communication, the importance of sound transparency is particularly prominent. It serves as the basic technical foundation for many important application fields such as medical ultrasound [1–5] and underwater acoustic communication [6–9]. However, current research focusing on underwater sound transparency remains largely unexplored. Achieving efficient sound transmission between solid and water is still an urgent challenge in acoustics. The inherent impedance mismatch between materials and water causes sound waves to reflect when passing through these two media. This results in a significant loss of energy and serious distortion of information transmission, affecting the imaging quality of medical ultrasound and hampering the efficiency of underwater

acoustic communication [6]. Therefore, fully exploring the potential of sound transparency in the underwater environment is crucial for theoretical research and practical applications.

The traditional approach to enhancing acoustic transmission is inspired by optical Fabry-Perot resonance [10–13], which necessitates intricate microcrack etching to facilitate sound wave reflections within resonant cavities [14,15]. Nevertheless, this intricate and highly precise process often results in structures with suboptimal mechanical properties, thereby limiting their widespread practical applications. To address these issues, researchers propose the utilization of transmission enhancement materials that integrate a single micropore or microcrack with a periodic grid [16–21]. Nevertheless, a common limitation of both structures is the requirement for an opening to connect two sides, which significantly limits their applicability in underwater sound transmission enhancement. The appearance of hydrogels [22,23] or pentamode materials [24] based on impedance matching provides a new way to realize hydroacoustic transparency in the case of flow isolation of water. However, they have issues with strength or complexity, respectively. The development of acoustic metamaterials [25–45] has significantly advanced the field of sound wave manipulation and introduced novel concepts for achieving sound transparency, offering unprecedented opportunities for innovation and progress in this domain. Given the significantly higher characteristic impedance of water compared to air, the rigid assumption for the thin walls of underwater metasurfaces becomes invalid

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in water [36–45]. In this case, the response system composed of a metasurface and fluid environment supports the coupling of the structural mode and sound mode. By designing metasurfaces based on this property, researchers have been able to achieve underwater sound transparency without any openings [46–52]. However, the frequency bands of these operations are very narrow, which limits the use of underwater sound-transparent metasurfaces. The ability of broadband sound waves to carry information and focus sound energy is unmatched by single-frequency sound waves. Thus, the pursuit of broadband sound transparency represents an inevitable trajectory in the evolution and advancement of sound transparency technology.

In this work, we break through the inherent limitations of the design of the sound-transparent metasurface and realize broadband underwater sound transparency spanning from 170 to 230 kHz based on fluid-structure interaction by multilayer structures. Leveraging the abundance of geometric parameters and the flexibility of degrees of freedom, we achieve tunable bandwidth manipulation, enabling the design of dual broadband transparency within the frequency ranges of 280–350 kHz and 450–510 kHz. The experimental results and simulation results show the same trend, which proves the validity of our proposed design. Our work offers vital insights for underwater acoustic communications and related ultrasonic medical imaging applications.

II. RESULTS AND DISCUSSION

A. Design of the broadband sound-transparent metasurfaces

Figure 1(a) illustrates a schematic of the acoustic metasurface designed to achieve broadband sound transparency. In order to effectively separate the flow of media on both sides of the structure, we choose the plate structure as the design theme [49,50]. Highlighted in the right panel of Fig. 1(a) are the intricate design details of the unit. Each unit features a middle layer (media isolation layer) with a thickness of t in the y direction, while its length in the x direction is fixed as the lattice constant P . The thickness of the other layers along the y direction is h_3 , and the distances from the middle layer are h_1 and h_2 , respectively. Additionally, the lengths of the other layers in the x direction are designated as w_1 and w_3 . The five-layer structure is coupled by a beam with a width of w_2 along the x direction. The design of the multilayer structure can introduce more local resonance modes into the structure [28], and the multimode coupling provides more freedom of adjustment for the metasurface. The symmetrical structure prevents the appearance of asymmetric transmission of waves and ensures the consistency of transmitted waves on both sides of the metasurface [25]. The detailed parameters of the metasurface are shown in Fig. 1.

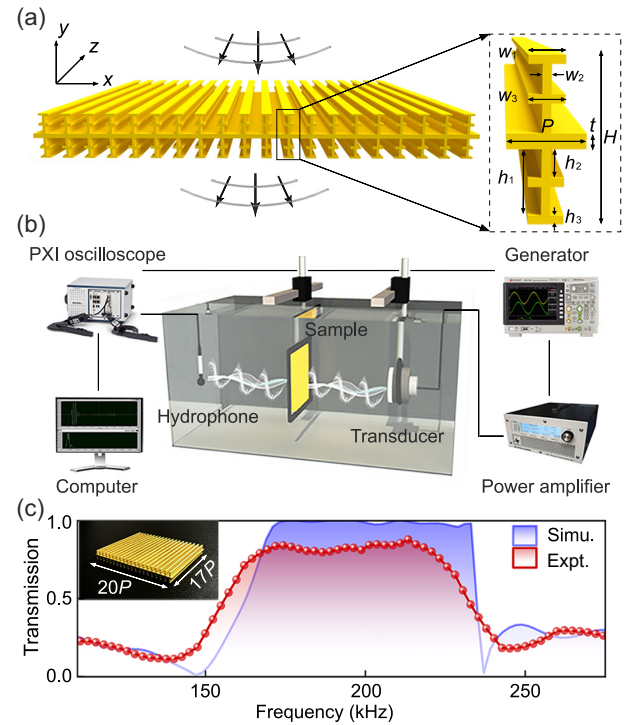


FIG. 1. (a) A conceptual three-dimensional visualization of the acoustic metasurface comprising the periodic brass unit is presented along with a parameter diagram outlining the dimensions of a single lattice: $P = 3.5$ mm, $t = 0.7$ mm, $w_1 = 1.6$ mm, $w_2 = 0.4$ mm, $w_3 = 1.6$ mm, $h_1 = 3$ mm, $h_2 = 1.3$ mm, $h_3 = 0.4$ mm, and $H = 7.5$ mm. (b) Experimental equipment diagram; the experimental sample is kept in a horizontal line with the center of the transducer and hydrophone. (c) Simulation (blue line) and measurement (red dots) results of transmission of the designed broadband sound-transparent metasurface. The illustration is a photograph of the experimental sample.

B. Numerical simulations and experimental demonstration of the sound-transparent metasurfaces

The acoustic properties of the proposed metasurfaces were verified using COMSOL Multiphysics. We set the density of water to 1000 kg/m^3 , the speed of sound to 1450 m/s , and specified the structural material as brass (density 7850 kg/m^3 , Young's modulus $205 \times 10^9 \text{ Pa}$, and Poisson's ratio 0.28) to simulate the acoustic performance of metasurfaces in practical applications. During the simulation, to fully evaluate the frequency characteristics of the metasurface under plane-wave incidence, we placed the designed unit inside a waveguide. The waveguide walls were treated as periodic boundaries, and the two ends of the waveguide were configured as perfectly matched layers. The incident end was set up for plane-wave incidence. Considering the interaction between fluid and solid we introduced the boundary conditions as follows:

$$\mathbf{F}_a = p_t \mathbf{n}, \quad (1)$$

$$-\mathbf{n} \cdot \left(-\frac{1}{\rho_c} \nabla p_t \right) = u_{tt}, \quad (2)$$

where u_{tt} is the structural acceleration, $\mathbf{n}$ is the surface normal, p_t is the total acoustic pressure, and $\mathbf{F}_a$ is the load borne by the structure.

In order to verify the accuracy of the simulation results, we also conducted corresponding experimental tests of the designed metasurfaces. The experimental equipment is shown in Fig. 1(b). The brass samples were fabricated with precision using computer numerical control technology, with a height of 60 mm and a width of 70 mm [as shown in Fig. 1(c)]. This dimensional design ensures that the ultrasonic transducer (Olympus V389-SU) can fully encompass the sample, enabling efficient measurement. During the experiments, a single-frequency signal was generated using a function generator (AFG3022C), emitting pulsed plane waves with a duration ranging from 2 to 5 cycles. Subsequently, to ensure sufficient signal strength for the experiment, a power amplifier (RF2100L) was employed to amplify the incident pulsed signal. To measure the acoustic pressure variations after the sound waves passed through the sample, we adopted a method similar to insertion loss measurement. Specifically, a hydrophone (Onda HNC-1500) was positioned precisely 10 cm away from the transducer, centered in the transducer's radiation path, to capture the transmitted acoustic signals under conditions of placed and unplaced samples. The received

acoustic pressure signals were then recorded and analyzed in real time using an oscilloscope (PXI 5124). By scanning through the frequency range and calculating the signal amplitudes, the acoustic transmission coefficients were obtained.

Figure 1(c) illustrates the transmission performance of the metasurface. The metasurface achieves broadband sound transmission enhancement in the 170 to 230 kHz band, which is in sharp contrast to the traditional narrow-band sound transmission enhancement structure in the frequency band characteristics. Overall, measurement results align well with simulation results, but discrepancies exist in transmission amplitude. These are mainly due to differences in the plane wave and Gaussian beam between simulation and experiment, sample fabrication, and intrinsic loss.

C. The working mechanism of the broadband sound-transparent metasurfaces

Essentially, the ability of sound waves to penetrate underwater solid structures depends on whether they evoke the inherent modes of fluid-solid systems. In order to further analyze the operation mechanism of our proposed metasurface, we conducted a detailed analysis from three dimensions: structural mode band structure, structural mode excitability, and system response considering fluid-structure interaction. In Fig. 2(a), we calculate the structural mode band structure (dots) of the metasurface using the eigenfrequency study module. For eigenfrequency simulation, the radiation loss of the structure is neglected. In the energy band analysis, Floquet periodic

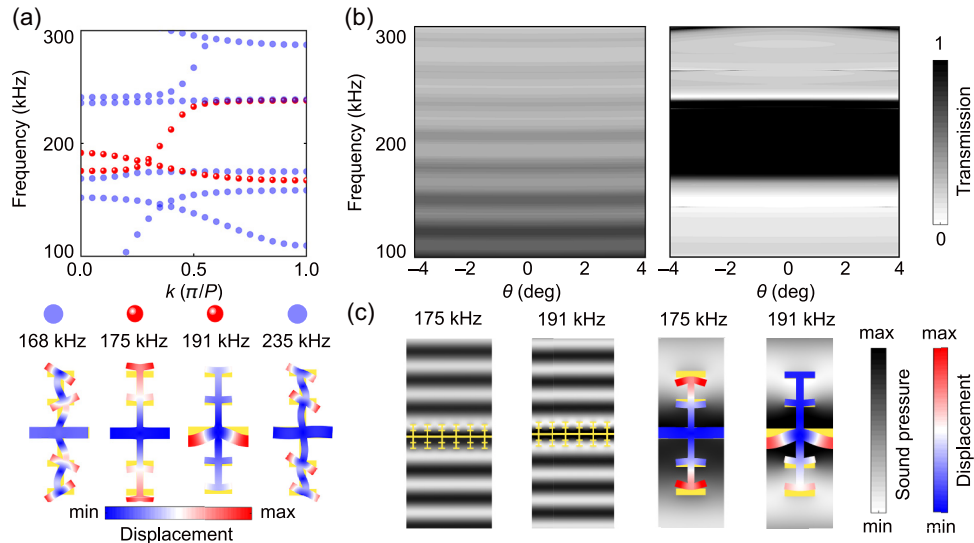


FIG. 2. (a) The figure presents the numerical dispersion curve of the structural modes. The modes involved in coupling are highlighted with red dots. The structural mode at the zero point, where $k = 0$, is also presented. (b) Angle-dependent transmission spectra for both the unit of the metasurface (right) and uniform brass plate (left). (c) Distributions of sound pressure intensity and total displacement of the metasurface and unit at 175 and 191 kHz. The color bar represents displacement, while the gray bar illustrates sound pressure in the water.

edge conditions are applied to the lattice along the periodic direction. Among them, the modes that support enhanced transmission are highlighted in red and are symmetric about the y axis at the zero point. This symmetry indicates that these modes can be efficiently excited by plane waves. In sharp contrast, the modes in blue (modes at 168 and 235 kHz) are asymmetric at the zero point and cannot be excited by plane waves. They are decoupled from the plane-wave modes, meaning that these structural modes do not participate in the metasurface's transmission behavior when the plane wave is incident. Figure 2(b) clearly shows the transmission response in relation to the excitation mode. The brass plate lacks a state of lossless transmission over the entire spectrum, which means that sound waves encounter obstacles when passing through the brass plate. However, when we observe how the metasurface behaves, the picture is quite different. In the frequency range aligned with excitable structural modes, the metasurface significantly enhances the transmission of sound waves. This is due to the hybridization of structural modes under the influence of fluid-structure interaction and near-field acoustic coupling, resulting in multiresonance coupling in the frequency band (multiple resonant responses occur in the system composed of the metasurface and surrounding water), and thus achieving complete transmission of sound waves. This discovery reveals the unique advantages of metasurfaces in the transmission of sound waves. Figure 2(c) shows the steady-state sound field and vibration position transformation of the metasurface and unit at corresponding frequencies. When the metasurface is at 175 kHz, the response mode of the system mainly inherits the structural mode of the metasurface at 175 kHz, and strong local resonance occurs on both sides of the plate. They jointly construct a virtual energy channel, so that the sound energy enters the transmitted end perfectly from the incident side. When the metasurface is at 191 kHz, the response mode of strong local resonance in the interlayer

is excited, which acts as an energy channel to make the sound energy perfectly enter the transmitted end from the incident side. The sound field intensity at the incident end and the transmitter end of the metasurface are basically the same, indicating that the sound wave is completely transmitted. The displacement field distribution of the structure is different under fluid-structure interaction coupling. This finding further confirms our conjecture that the enhanced transmission of sound waves in the corresponding frequency band is caused by the coupling of multiple response modes.

D. Customizing the transmitted acoustic enhancement

The working mechanism of multimode coupling plays a crucial role in fine-tuning the metasurface's transmission response. By precisely controlling the coupling strength and natural frequency between the layers, we can effectively manipulate the metasurface's properties. We select three key structural parameters for further study: w_1 , which determines the natural frequency of the upper and lower layers; h_2 and w_2 , both of which regulate the coupling strength between the layers. The transmission line corresponding to the selected sample structural parameters is highlighted by the blue curve in Fig. 3. It can be clearly observed from the figure that even when the internal parameters of the metasurface undergo slight changes, the transmission spectrum remains stable, without any notable performance degradation or failure. This ensures the consistent effectiveness of the underwater sound transmission enhancement performance. This demonstrates that the metasurface is robust against manufacturing tolerances, thus providing strong assurance for its stable and reliable performance in practical applications, enabling it to exhibit excellent performance under various working conditions. By further comparing the changes of different parameters in Fig. 3, we can find that the changes in these structural

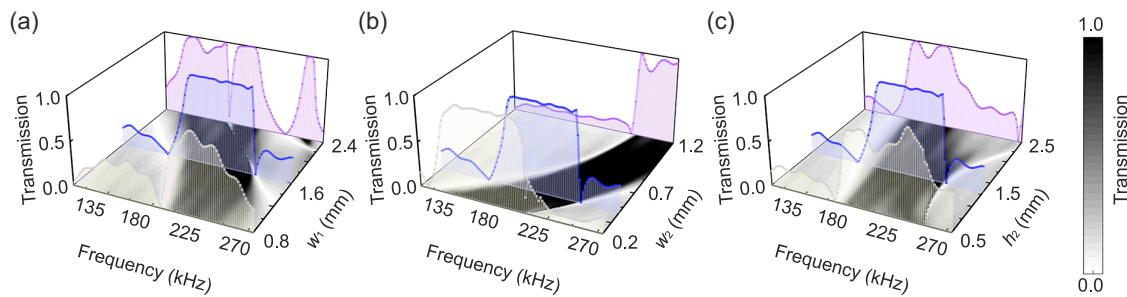


FIG. 3. This figure demonstrates the influence of varying structural parameters on the transmission spectrogram. Specifically, the following three parameters are analyzed in detail: (a) the width of the top and bottom cover plate w_1 , (b) the width of the beam rib w_2 , and (c) the height of the secondary beam rib h_2 . The color map visually represents the changes in transmission, with darker shades indicating more efficient sound energy transmission. The longitudinal cross-sectional view emphasizes the transmission trend varying with frequency under fixed parameters, and specifically illustrates the transmission characteristics for the selected structural parameters, with the blue segment emphasizing the optimal design parameters employed in this study. The structural parameters of the design, excluding the tuning variables in each figure, are fixed at the following values: $w_1 = 1.6$ mm, $w_2 = 0.4$ mm, and $h_2 = 1.3$ mm.

parameters have a significant impact on the operating bandwidth and frequency characteristics. With adjustments to the values of w_1 and h_2 , the working bandwidth shows an obvious separation and narrowing trend. For example, as shown in Fig. 3(a), with the increase of w_1 , the originally continuous transmission spectral lines are gradually separated into multiple independent peaks, forming a comblike structure with multiple passbands. w_2 mainly affects the operating frequency range of the metasurface. As shown in Fig. 3(b), with the increase of w_2 , although the width of the operating bandwidth does not change significantly, the center frequency of the bandwidth shifts significantly to the high-frequency direction. This proves that, as a coupling bridge between layers, the connecting beam is an important part of the metasurface, and changing it will affect the coupling effect between the whole metasurface, and then change the working frequency of the metasurface. These parameter studies finally prove that the proposed metasurface has excellent tunability, enabling us to effectively adjust the transmission enhancement to optimize metasurface performance by precisely controlling these parameters.

E. Dual broadband sound transparency design

In the previous section, we verified the tunability of acoustic metasurfaces based on parametric analysis. Based on research into the parameter adjustment, the structural parameters were precisely tuned, enabling successful manipulation of the resonance mode frequency.

Subsequently, this leads to the creation of an acoustic metasurface design featuring dual broadband characteristics. Compared with the single broadband design, the information-carrying capacity of the dual broadband channel is significantly improved, and the doubling effect is achieved. The specific parameters used in this design are shown in Fig. 4(a). As can be seen from the specific parameters given in the figure, during the design process we kept the thickness of the middle layer constant, while adjusting other parameters to affect the location of its operating frequency and bandwidth, so as to achieve dual broadband sound transmission for underwater communication equipment. The dispersion curve of the structural modes involved in the transmission is highlighted on the left side of Fig. 4(b). The adjustment of structural parameters distributes the modes in two frequency bands. Correspondingly, when fluid-structure interaction and radiation are considered, the hybrid system response appears as two separate transmission bands. It is worth noting that in between the two bands we can clearly observe the bound state in the continuum [53,54] composed of the structure and the fluid system (the transmittance width around 400 kHz gradually disappears with decreasing angle). The plane-wave mode is decoupled from the mode response, and when the angle of the incident wave is deflected, the remaining structural modes and the sound field modes are coupled with the incident wave. This angle-sensitive mode provides a new design concept for sensitive underwater sensors. Overall, the design not only improves the efficiency of

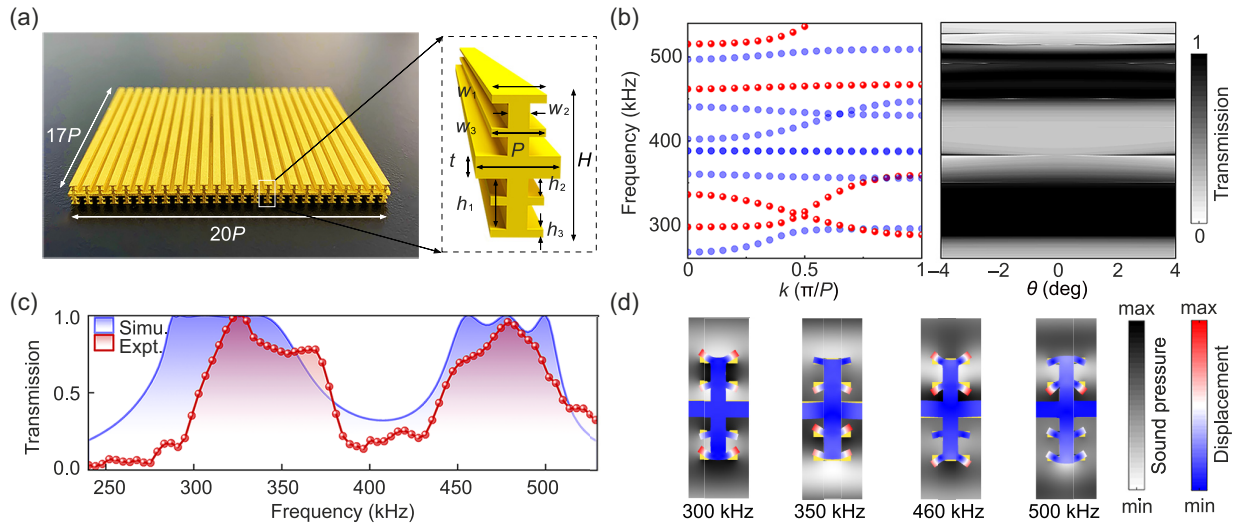


FIG. 4. (a) This figure illustrates the experimental sample of dual broadband design (on the left) alongside the optimized parameter values for the unit (on the right). The specific parameters include $P = 2.56$ mm, $t = 0.7$ mm, $w_1 = 1.62$ mm, $w_2 = 0.64$ mm, $w_3 = 1.62$ mm, $h_1 = 1.6$ mm, $h_2 = 0.58$ mm, and $h_3 = 0.26$ mm. (b) The figure presents the numerical dispersion curve of the structural modes. The modes involved in coupling are highlighted with red dots (left). Angle-dependent transmission spectra for the unit of the metasurface (right). (c) Simulation (blue line) and measurement (red dots) transmission of the designed dual broadband sound-transparent metasurface. (d) Spatial distributions of sound pressure intensity and total displacement are visualized for the individual cell at 300, 350, 460, and 500 kHz. The color bar represents displacement in the unit, while the gray bar illustrates sound pressure in the water.

information transmission but also demonstrates the great potential of acoustic metasurfaces in the field of communication and sensing. Figure 4(c) shows the comparison of simulation results and experimental results. Although there are amplitude and frequency deviations caused by differences in experimental equipment and sample production, both experimental results and simulation results show clear dual-passband spectrum characteristics, which fully verify the effectiveness of the design. Combined with the steady-state structural mode given in Fig. 4(d), we can conclude that the formation of the double broadband phenomenon is caused by the cooperation of multiple modes. The proposed dual broadband design realizes the physical independence of multiple receiving channels and provides an alternative concept for design and application.

III. CONCLUSIONS

In summary, we present an acoustic metasurface that offers enhanced transmission capabilities through multi-mode resonance coordination under fluid-solid interaction. This metasurface overcomes the limitations of narrow-band effectiveness, enabling effective broadband sound transparency. The metasurface's rich structural parameters enable band-editing functionality, aiding in versatile wave manipulation. Simulation and experimental results align, effectively validating our proposed construction approach. This metasurface achieves broadband sound transparency spanning from 170 to 230 kHz, with dual broadband sound transparency offering passbands of 280–350 kHz and 450–510 kHz. Our design method and metasurface transcend the single-frequency constraints of conventional sound transmission enhancement metasurfaces, offering vital insights for underwater acoustic detection and related ultrasonic medical imaging applications. More significantly, our research stands as a stepping stone in the exploration of broadband metasurfaces, opening up an alternative path for advancements in this research field.

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